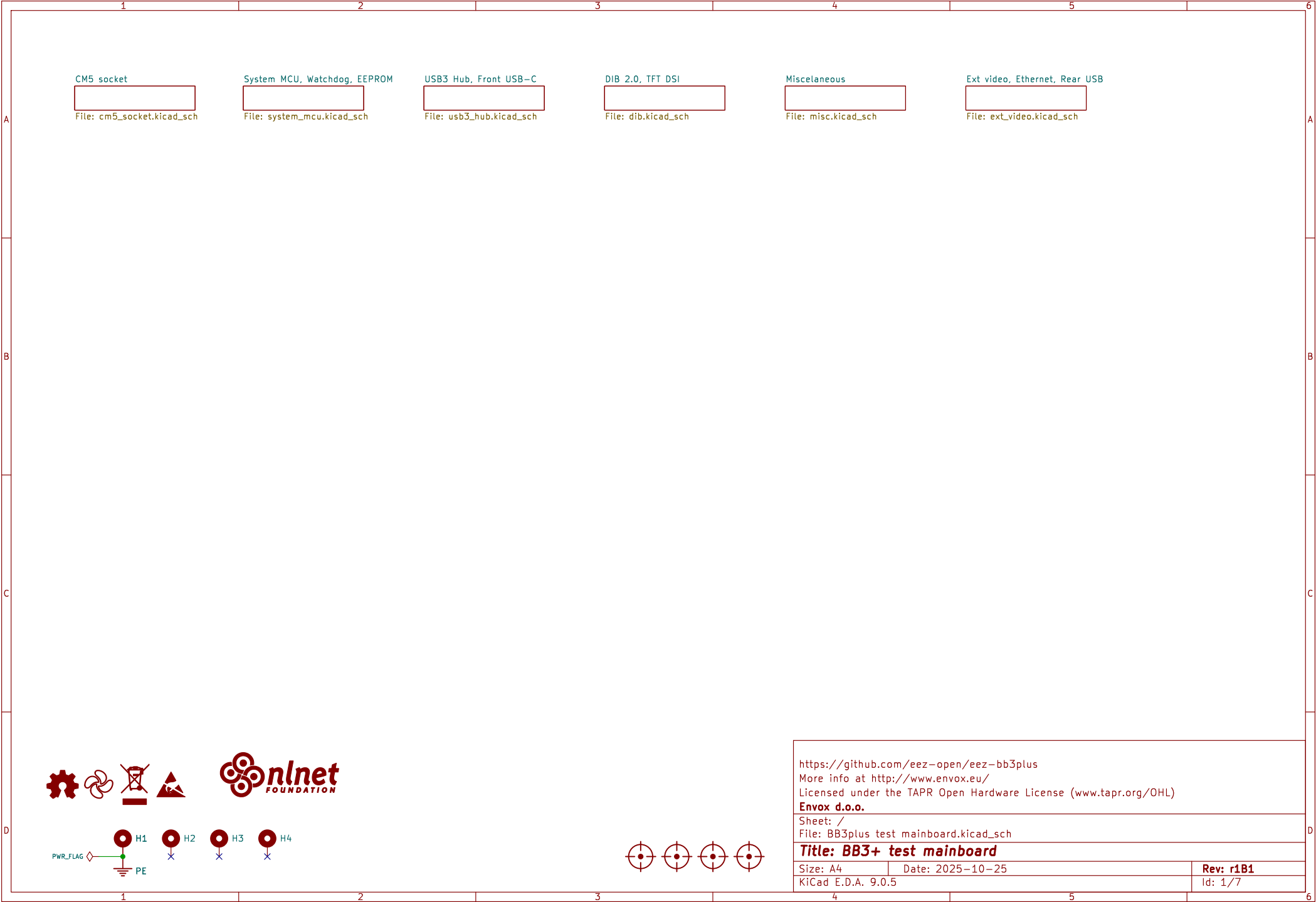
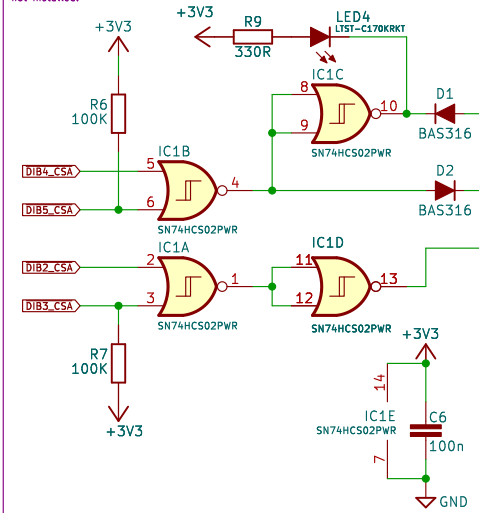


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Sheet: / File: BB3plus test mainboard.kicad_sch				
Title: BB3+ test mainboard				
Size: A4	Date: 2025-10-25	Rev: r1B1		
KiCad E.D.A. 9.0.5	Id: 1/7			

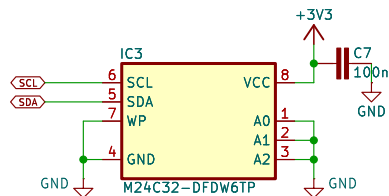


System 32-bit MCU

The System MCU is programmed so that the Master MCU first simultaneously activates DIB4_CSA and DIB5_CSA and then simultaneously activates DIB2_CSA and DIB3_CSA which will reset the System MCU and put it into bootloader mode. Simultaneous activation of two CSA (Chip Select) signals on the same SPI channel should never occur in normal operation. 100K pullup resistors ensure that RESET and BOOT0 are inactive if the Master MCU is not installed.

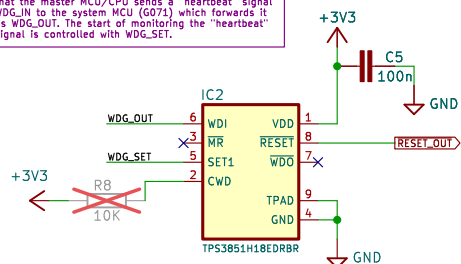


EEPROM

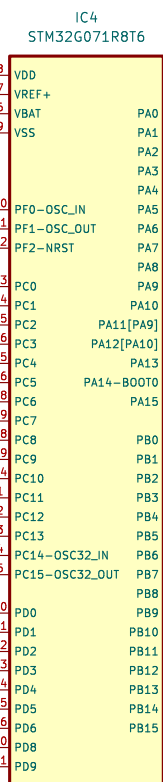
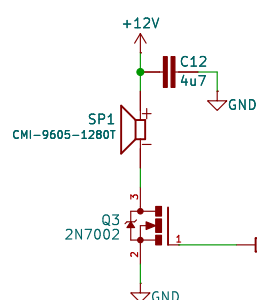


Watchdog/supervisor

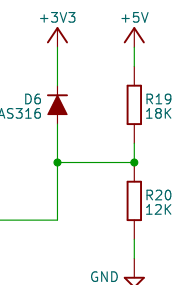
The watchdog signal is daisy chained in such a way that the master MCU/CPU sends a "heartbeat" signal WDG_IN to the system MCU (G071) which forwards it as WDG_OUT. The start of monitoring the "heartbeat" signal is controlled with WDG_SET.



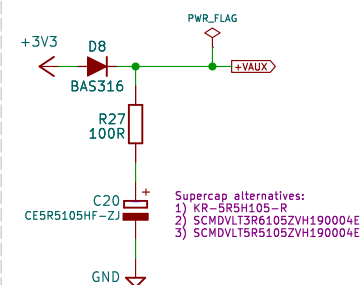
Audio out



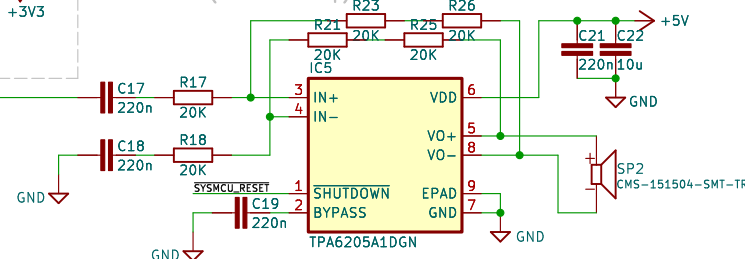
5V monitoring



Supercap for RTC



Audio out (AB-class amp)



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Sheet: /System MCU, Watchdog, EEPROM/

File: system_mcu.kicad_sch

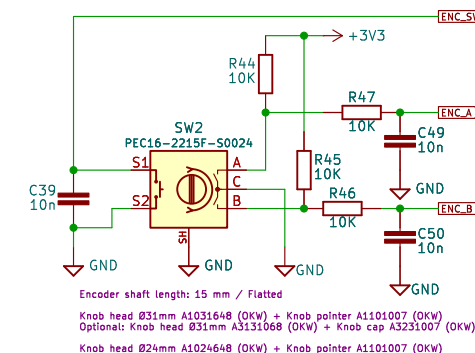
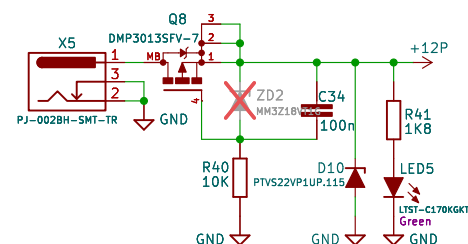
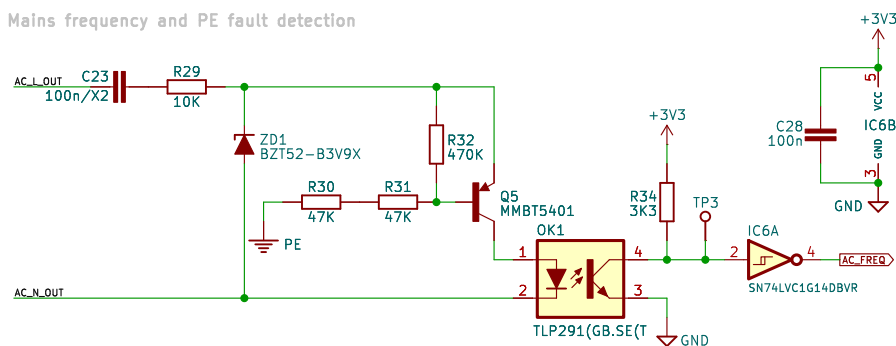
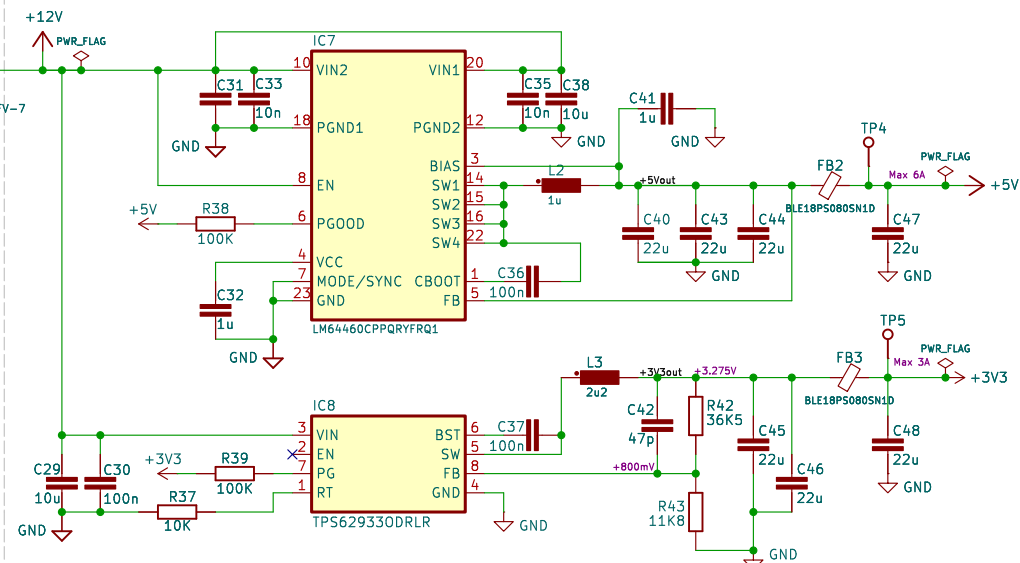
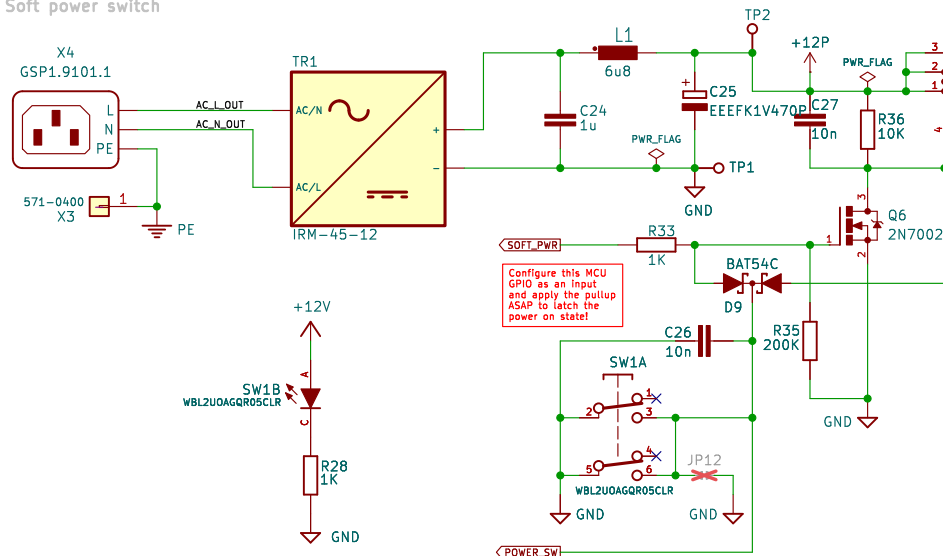
Title: BB3+ test mainboard

Size: A4 Date: 2025-10-25

KiCad E.D.A. 9.0.5

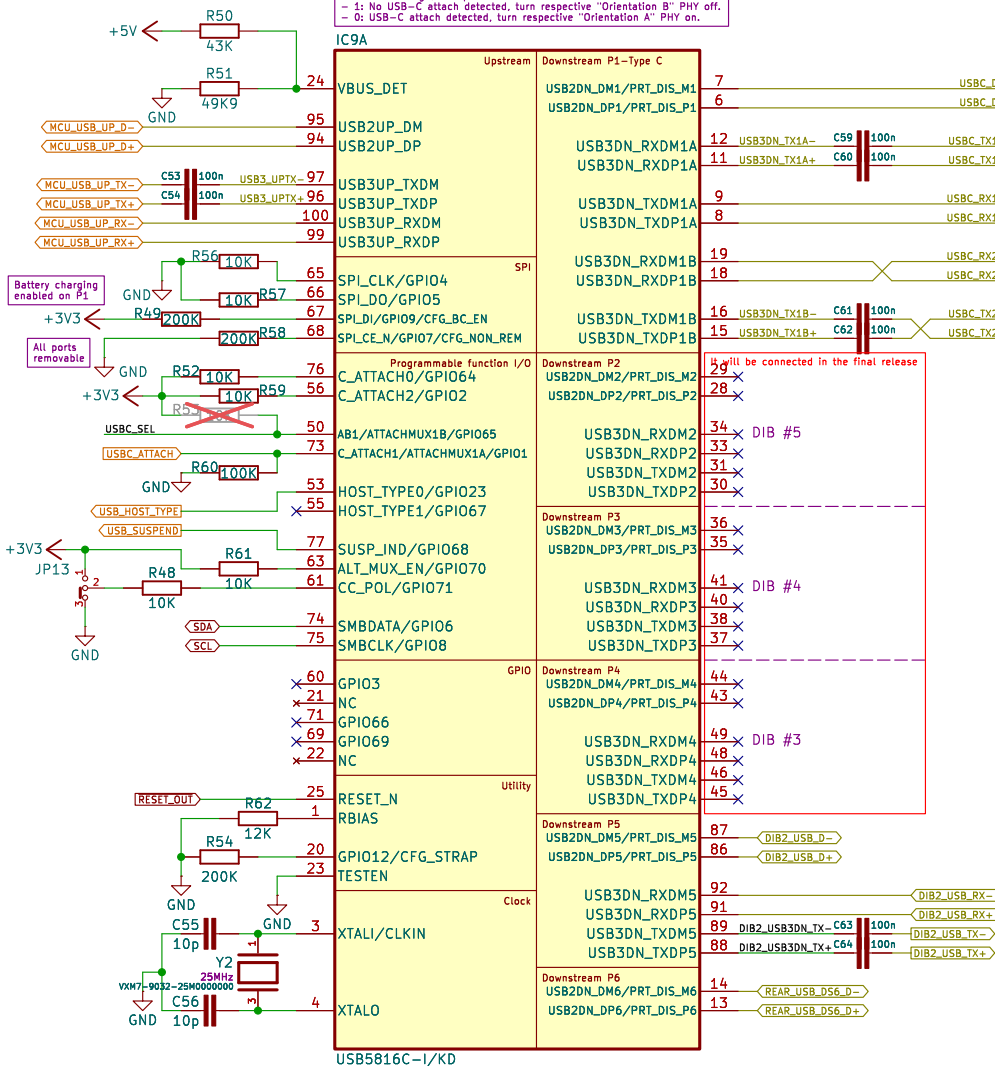
Rev: r1B1

Id: 3/7

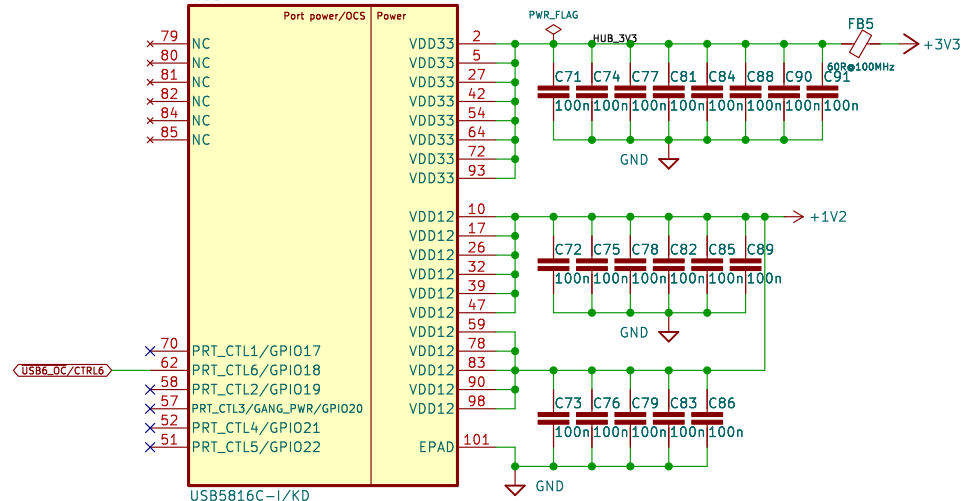


Id: 4/7

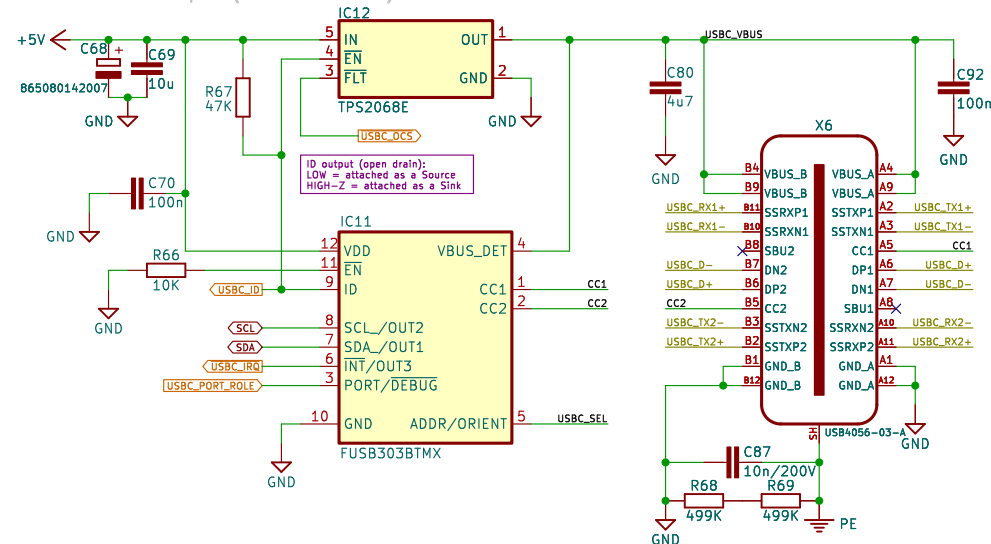
6-port USB hub



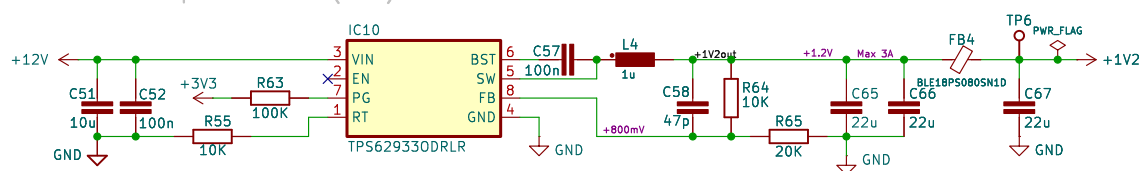
IC9B



USB-C 3.0 front port (source max 1.5A)



USB hub DC-DC stepdown converter (+1V2)



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Sheet: /USB3 Hub, Front USB-C/

File: usb3_hub.kicad_sch

Title: BB3+ test mainboard

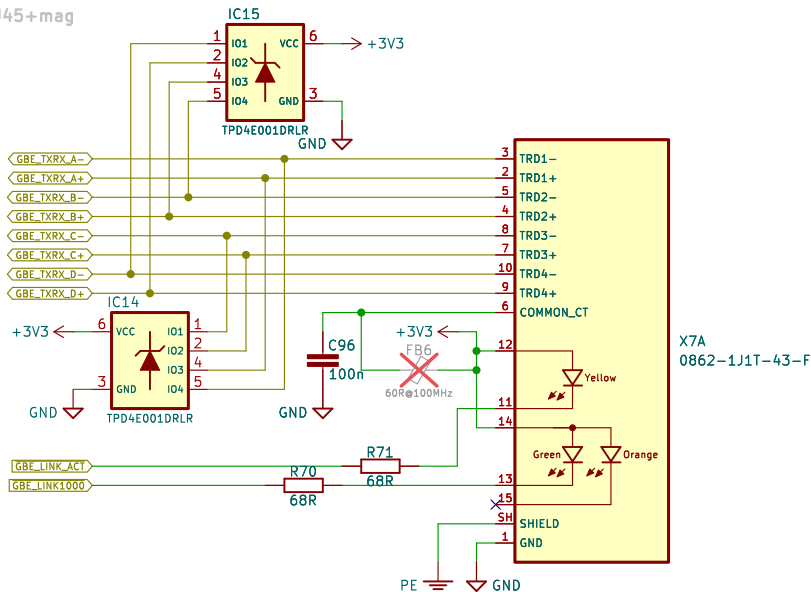
Size: A4	Date: 2025-10-25
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Size: A4	
KiCad E.D.A. 9.0.5	

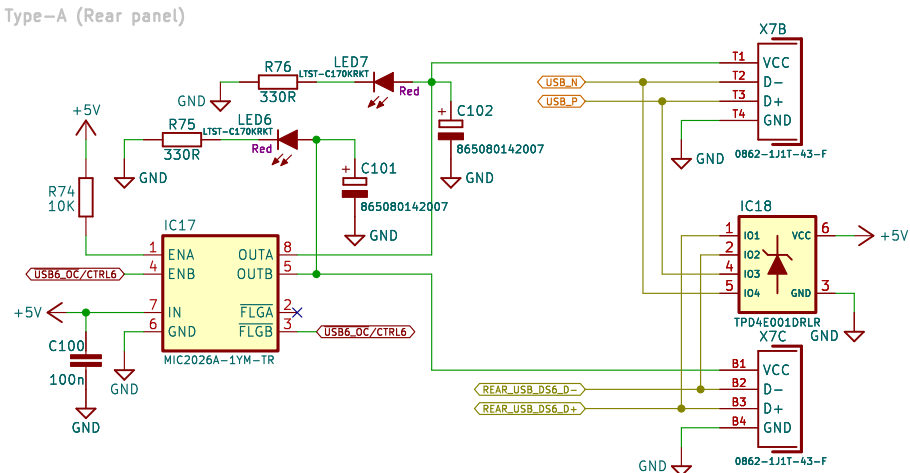
Rev: r1B1

Id: 5/7

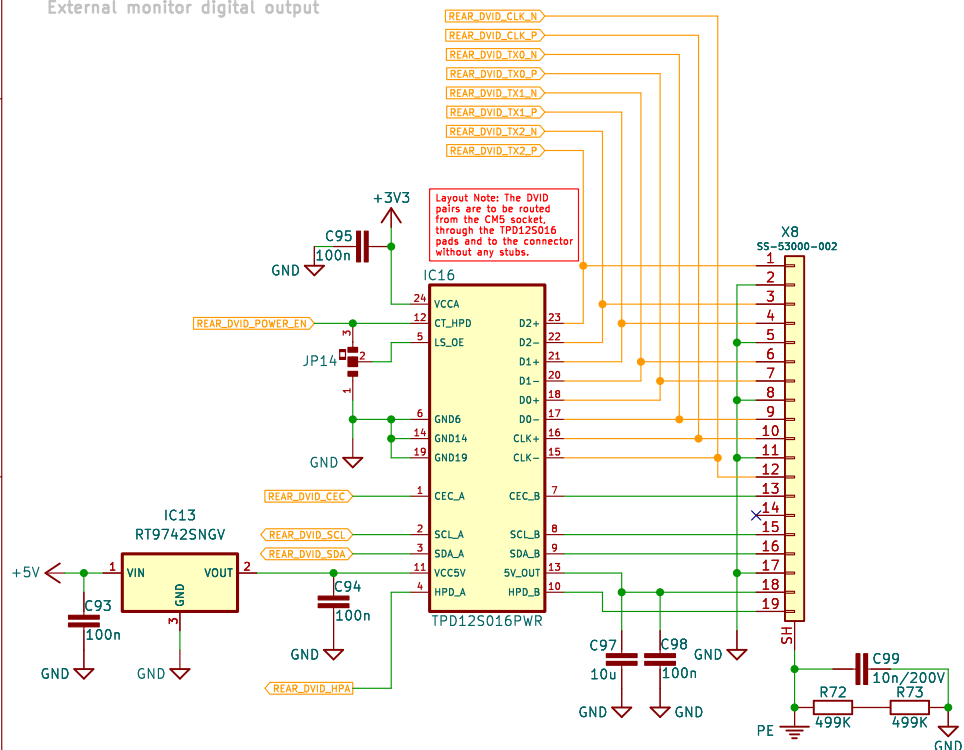
Ethernet RJ45+mag



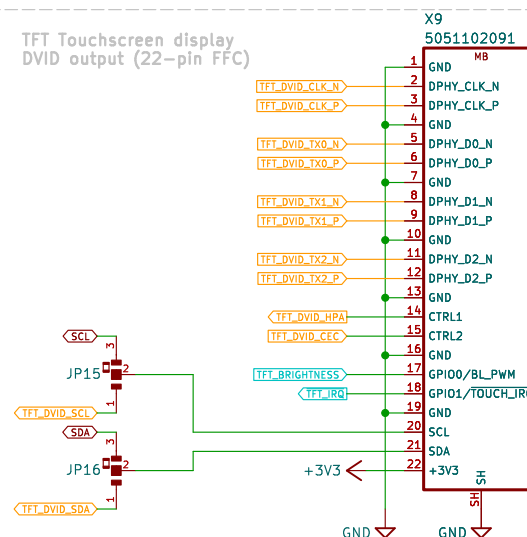
USB2.0 Type-A (Rear panel)



External monitor digital output



TFT Touchscreen display DVID output (22-pin FFC)



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Envox d.o.o.

Sheet: /Ext video, Ethernet, Rear USB/

File: ext_video.kicad_sch

Title: BB3+ test mainboard

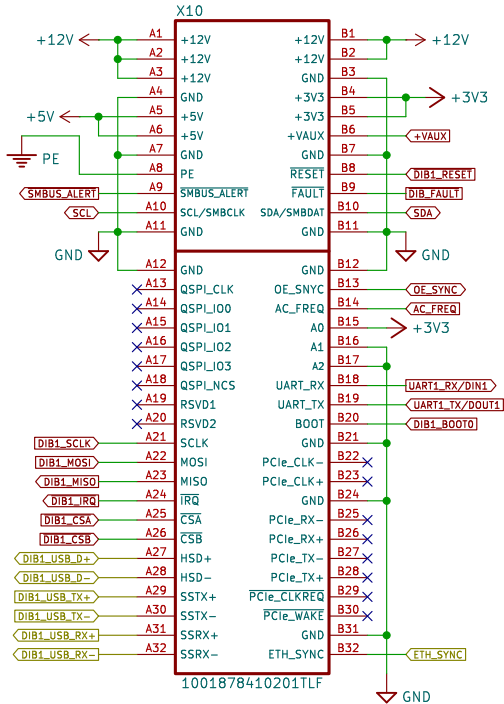
Size: A4 Date: 2025-10-25

KiCad E.D.A. 9.0.5

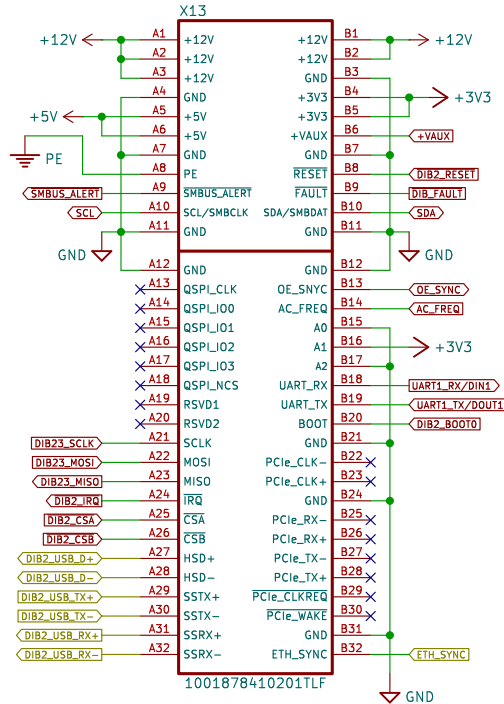
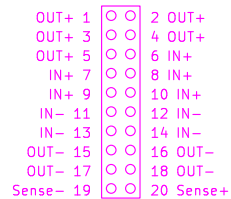
Rev: r1B1

Id: 6/7

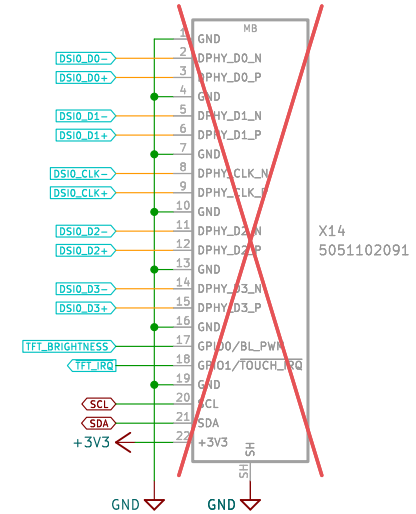
DIB 2.0 Signal connectors



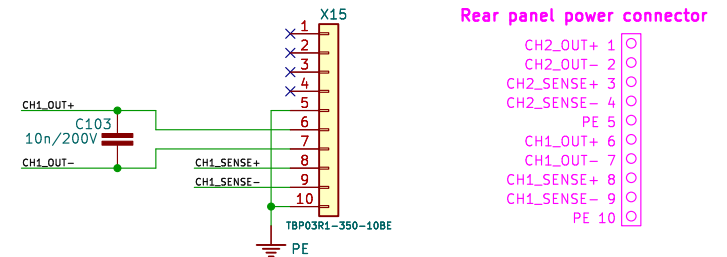
20-pin DIB v2.0 Power connector



TFT Touchscreen display DSI output (22-pin FFC)



Rear front power outputs



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Sheet: /DIB 2.0, TFT DSI/
 File: dib.kicad_sch

Title: BB3+ test mainboard

Size: A4 Date: 2025-10-25

KiCad E.D.A. 9.0.5

Rev: r1B1

Id: 7/7